

Digital Integrated Circuit Design

HW-3: Memory Read/Write Control using FSM

Problem:

Based on HW2, please design an FSM to do read/write control for accessing memory.

- 1.1 Both state-table and state-diagram are described in Table 1 and Fig. 2, respectively. The **SelfTST** (Self test) is used to enable the memory access control.

Table 1. State table and state diagram

Input		Internal Input		Current State	Output	
RESET_	SelfTST	WRENAB_Done	RDENAB_Done		WENAB_FSM	RENAB_FSM
0	X	X	X	ST0	1	0
1	0	0	0	ST0	1	0
1	1	0	0	ST1	0	0
1	1	1	0	ST2	1	1
1	1	1	1	ST3	1	0

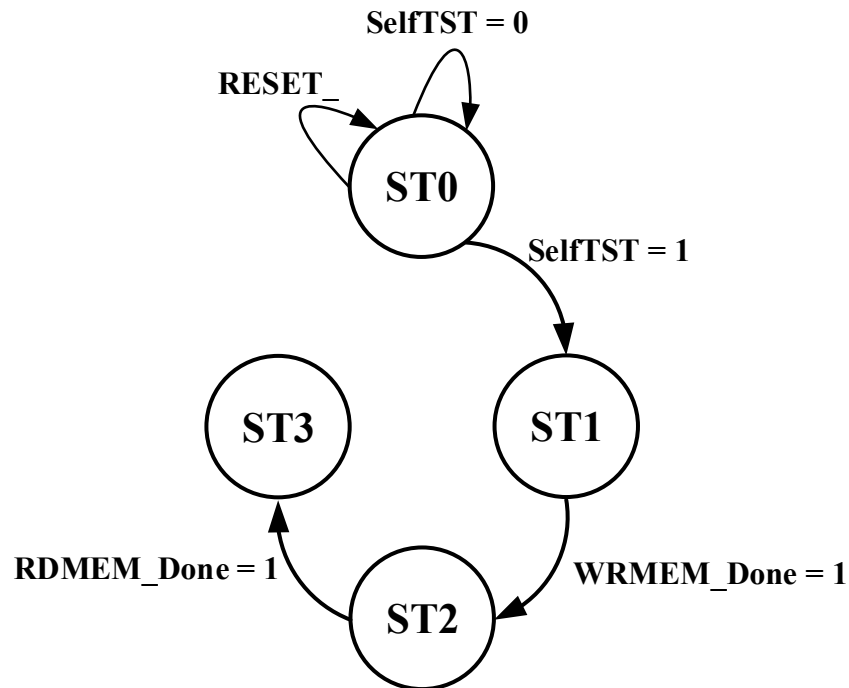


Fig. 1. Stage diagram.

- 1.2 The block diagram of the memory read/write access control using FSM is shown in Fig. 2. All of signals, **Data_In** (the data input of memory), **Data_Out** (the data output of memory), **CLK**, and **RESET_**, are provided in testbench, **HW3_MEM_FSM_rtb.v**. Besides, **WENAB_FSM** (the write enable with active “Lo”) and **RENAB** (the read enable with active “Hi”) are **generated from FSM**. In order to change the current state (from ST1 to ST2 and from ST2 to ST3), **WRMEM_Done** (Write operation done) and **RDMEM_Done** (Read operation done) must be feedback to FSM to do the related operations. Both **WRMEM_Done** and **RDMEM_Done** indicate that the **write** and the **read** operations are completely finished, respectively.

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module name: MEM_FSM

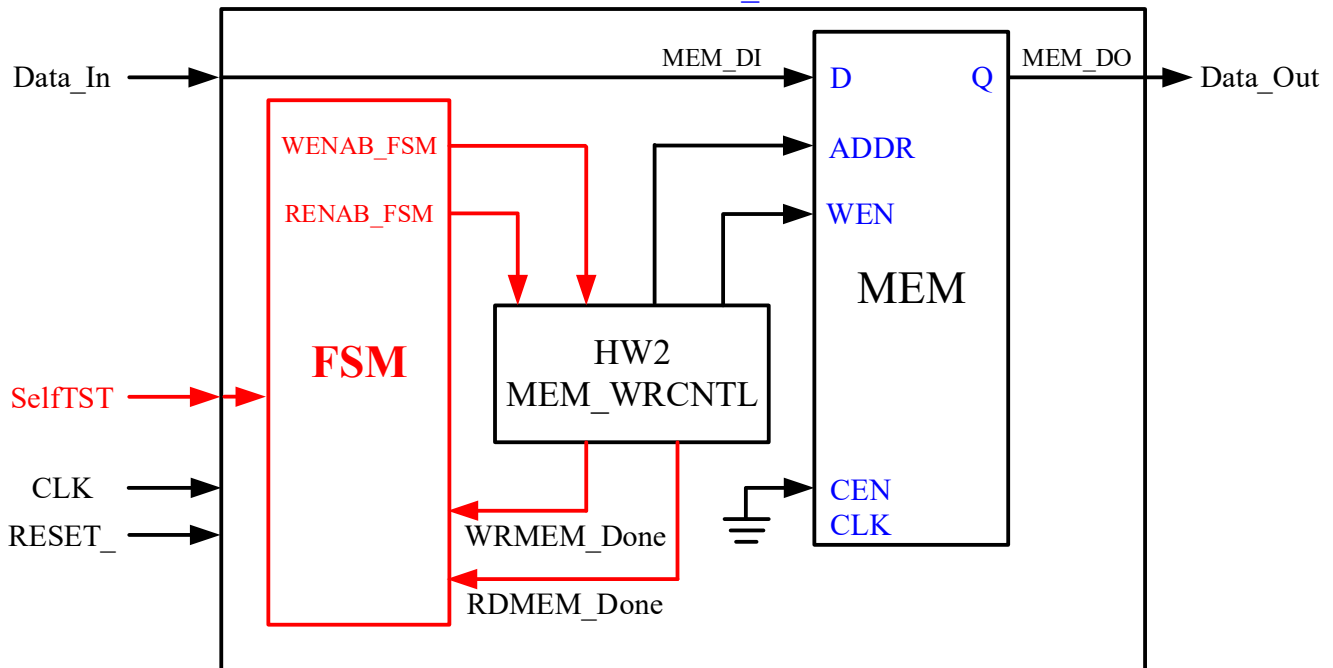


Fig. 2. Block diagram of the memory read/write access control using FSM

- 1.3 The template Verilog RTL **HW3_MEM_FSM_rtb.v** and **HW3_FSM_CNTL_rtl.v** are also uploaded in E-learning. The testbench **HW3_MEM_FSM_rtb.v** is used to evaluate your designs including **HW3_FSM_CNTL_rtl.v** and your **HW2** design.
- 1.4 All of the **registers** in your design are assumed to operate in the **rising** edge-triggered.
- 1.5 The related write timing waveform is shown in Fig. 3.
- 1.6 The related read timing waveform is illustrated in Fig. 4.
- 1.7 Please design a read/write control to access RF1_32x25b (**RF1_32x25b.v**) based on a given testbench. Of course, you can try to modify the testbench to evaluate your RTL code.

Submission

1. Please submit the following files to E-learning.
 - (a) **B123456_HW3_FSM_CNTL_rtl.v**: It is your **FSM** design, where **B123456** is your **student ID**.
 - (b) **B123456_HW3_MEM_WRCNTL_rtl.v**: The file is your **HW2** design.
 - (c) **B123456_HW3_MEM_FSM_rtb.v**: A testbench for you top-module design
2. Please deliver the simulation results and the circuit diagrams of your design (.docx) in class.
 - (a) For the simulation results, please attach the related photos (.jpg or pdf) in .docx.
 - (b) For the circuit diagrams, please sketch it using VISIO (.vsdx) or pen and then scan it (.jpg or .pdf). After that, please paste the circuit diagram in .docx.

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Fig. 3. Write waveform



Fig. 4. Read waveform

